

# JACKSON ESHBAUGH

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Bethlehem, PA — 18020, USA

## EDUCATION

### • Lafayette College

*B.S., Computer Science*

◦ GPA: 4.00/4.00

*August 2023–May 2027*

Easton, PA

### • Lafayette College

*A.B., French Language*

◦ GPA: 4.00/4.00

*August 2023–May 2027*

Easton, PA

## RESEARCH INTERESTS

### Topics

Computational Linguistics   Machine Learning Interpretability  
Theoretical Foundations of Artificial Intelligence

### Guiding Questions

How do linguistic structure and semantics interact in multilingual NLP models?

How does language work—in computers and in humans?

How do we make neural networks more transparent?

## PUBLICATIONS & PREPRINTS

C=CONFERENCE, J=JOURNAL, P=PREPRINT, S=IN SUBMISSION, T=THESIS

- S.2**   **Eshbaugh, J.**, Tiwari, C., and Silveyra, J. [A Modular and Multimodal Generative AI Framework for Urban Building Energy Data: Generating Synthetic Homes](#). 2026. Submitted to *Energy & Buildings*. [arXiv:2509.09794](#).
- S.1**   **Eshbaugh, J.** [Fidelity Isn't Accuracy: When Linearly Decodable Functions Fail to Match the Ground Truth](#). 2026. Submitted to the *Workshop on Scientific Methods for Understanding Deep Learning* at ICLR 2026 [arXiv:2506.12176](#).

## POSTERS & PRESENTATIONS

P=POSTER, T=TALK, K=KEYNOTE

- P.3**   **Eshbaugh, J.** (Dec 2025). "Synthetic Homes: A Modular and Multimodal Generative AI Framework for Urban Building Energy Data." *Lafayette College EXCEL Scholars Poster Session*, Easton, PA. [[poster](#)]
- P.2**   **Eshbaugh, J.** (Oct 2025). "Synthetic Homes: A Modular and Multimodal Generative AI Framework for Urban Building Energy Data." *CyberAccelerate Poster Session at KINBERCON 2025*, Lancaster, PA. [[poster](#)]
- P.1**   **Eshbaugh, J.** (Sep 2025). "Synthetic Homes: A Modular and Multimodal Generative AI Framework for Urban Building Energy Data." *Lafayette College Bicentennial Weekend Poster Session*, Easton, PA. [[poster](#)]

### Upcoming & Invited Presentations

- T.1**   Silveyra, J. and **Eshbaugh, J.** (March 2026). "Synthetic Homes: A Modular and Multimodal Generative AI Framework for Urban Building Energy Data." *Invited talk at Georgia State University*, Atlanta, GA.

## RESEARCH EXPERIENCE

### • Independent Study—Linguistics of Idioms

*Department of Language and Literary Studies, Lafayette College*

◦ Advisor: Dr. Maria Hernandez

◦ Developing a theoretical model of idiomatic language, specifically for French

◦ Applying the model to idioms and interpreting how humans process idiomatic expressions

◦ Crafting a journal article in French that shares findings and implications

◦ This work will contribute the linguistic approach to my computational linguistics thesis.

*January 2026 - May 2026*

Easton, PA

### • Honors Thesis Research—Computational Linguistics

*Departments of Computer Science & Language and Literary Studies, Lafayette College*

◦ Advisors: Dr. Sofia Serrano (Computer Science) and Dr. Maria Hernandez (French)

◦ Developing FRIdiom: an annotated corpus for detecting idiomatic expressions in French

*February 2025 - Present*

Easton, PA

- Applying back-translation and neural methods to improve figurative language handling in multilingual MT
- Using interpretability techniques from prior work to probe language model representations
- Connecting linguistic and computational methods to determine how humans and computers process idiomatic language
- Dataset paper in preparation for submission

#### • **Independent Research—Neural Network Interpretability**

April 2025 - Present

Department of Computer Science, Lafayette College

Easton, PA

- Investigated when linear models fail to faithfully represent neural networks
- Designed experimental framework comparing surrogate fidelity vs. task accuracy across regression tasks
- Proposed the  $\lambda$ -score as a diagnostic metric; demonstrated high surrogate fidelity does not imply accuracy
- Implemented full ML pipeline: network training, surrogate evaluation, statistical analysis
- Manuscript reviewed by Dr. Jorge Silveyra and Dr. Jeffrey Pfaffmann prior to submission
- Result: Preprint “Fidelity Isn’t Accuracy” (arXiv:2506.12176).

#### • **EXCEL Scholar—Building Energy & Generative AI**

June 2025 - Present

Department of Computer Science, Lafayette College 

Easton, PA

- Advisor: Dr. Jorge Silveyra (Lafayette College)
- Collaborator: Dr. Chetan Tiwari (Georgia State University)
- Developed neural network approach to recommend energy efficiency retrofits at neighborhood scale
- Used EnergyPlus™ simulations and generative AI for urban building energy modeling
- Co-authored manuscript submitted to *Energy & Buildings* (arXiv:2509.09794)
- Presented research at KINBERCON 2025 and Lafayette poster sessions
- Invited to present at Georgia State University

#### • **ACL Manuscript Review**

October 2025 - Present

Department of Computer Science, Lafayette College

Easton, PA

- Collaborated as a secondary reviewer with Dr. Sofia Serrano on peer review of ACL Rolling Review manuscripts
- Evaluated research methodology, experimental design, and validity of scientific claims
- Drafted preliminary reviews for each submission which were refined through discussion with Dr. Serrano

## TEACHING EXPERIENCE

#### • **Department of Computer Science, Lafayette College**

August 2024 - Present

Teaching Assistant

Easton, PA

- Assist professors with teaching duties by helping students during class meetings.
- Praised by students as an outstanding teacher who cares and can explain complex things in ways that make sense.
- More information on my [teaching page](#).

#### • **Academic Resource Hub, Lafayette College**

August 2024 - Present

Mentored Study Group Leader

Easton, PA

- Hold two review sessions per week, covering course content.
- Produce materials such as worksheets and presentations for sessions.
- Praised by students as an outstanding teacher who cares and can explain complex things in ways that make sense.
- More information on my [teaching page](#).

## SKILLS

- **Programming Languages:** Python, Java, JavaScript, C
- **Machine Learning:** PyTorch, TensorFlow, NumPy
- **Other Tools & Technologies:** L<sup>A</sup>T<sub>E</sub>X

## HONORS AND AWARDS

#### • **Marquis Scholarship**

May 2023

Lafayette College



- Demonstrated intellectual curiosity in addition to superior academic achievement.
- Recipients display the Marquis’ ideals of global citizenship, scholarly pursuits, leadership, and concern for others.

#### • **Dean’s List**

F23, S24, F24, S25, F25

Lafayette College

- Achieved a Semester GPA of 3.60 or higher.

VOLUNTEER EXPERIENCE

- **Audio/Visual Specialist**  
*Hope Alliance Church*
    - Design and operate weekly worship slides and sermon visuals
    - Configure live stream and projection systems for services
    - Run slides and broadcast during live events
    - Troubleshoot technical issues quickly and independently
    - Maintain and reset the worship space after services
    - Apply skills in graphic design, live sound, and audiovisual tech
    - Serve faithfully by supporting worship through technical service
  - **Audio/Visual Specialist**  
*Lafayette DiscipleMakers Christian Fellowship*
    - Similar service to the above with no live stream component

July 2021 - Present


- September 2023 - Present



CERTIFICATIONS

- **Deep Learning Specialization**

June 2025

ADDITIONAL INFORMATION

Languages Spoken

English (Native Tongue), French (Fluent; Professional Competency)

Interests

Music (Jazz Vocals, Trombone, Piano, Composition); Photography and Photo Editing; Videography and Video Editing